

Ensemble Quantum Computing Laborator Prelab 0

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Values and dimensions of relevance to NMR

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- a) Magnetic moment of a Dirac electron:

$$\mu_{Dirac} = \frac{e\hbar}{2m} = 9.274010 \times 10^{-27} \text{ J/T}.$$

$$1 \text{ eV} = 1.602176634 \times 10^{-19} \text{ J}.$$

$$\mu_{Dirac} = 5.788 \times 10^{-8} \text{ eV}.$$

- b) The current (CODATA-recommended) best-measured value for the magnetic moment of an electron is $9.2848 \times 10^{-27} \text{ J/T} = 5.79 \times 10^{-8} \text{ eV/T}$
- c) A simple expectation of the magnetic moment of a proton using the same method is:

$$\mu_{Proton} = \frac{e\hbar}{2m_{Proton}} = 5.0508 \times 10^{-27} \text{ J/T} = 3.152 \times 10^{-8} \text{ eV/T}$$

- d) However the measured value is very different:

$$\mu_{pmeasured} = 1.4106 \times 10^{-26} \text{ J/T} = 8.8030 \times 10^{-8} \text{ eV/T}$$

Which is 2.7 times larger the naive expectation, the reason to this is the g factor of a proton is approximately 5.585, which is way bigger than 2.

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- a) Using equation $\gamma = \frac{2\mu}{\hbar}$, The gyromagnetic ratio of the expected magnetic moment of Dirac electron is $1.7589 \times 10^8 \text{ rad} \cdot \text{s}^{-1} \cdot \text{T}^{-1}$.

The gyromagnetic ratio of the measured magnetic moment of Dirac electron is $1.7609 \times 10^8 \text{ rad} \cdot \text{s}^{-1} \cdot \text{T}^{-1}$.

The gyromagnetic ratio of the proton(expected) is $9.5944 \times 10^7 \text{ rad} \cdot \text{s}^{-1} \cdot \text{T}^{-1}$.

The gyromagnetic ratio of the proton(measured) is $2.6753 \times 10^8 \text{ rad} \cdot \text{s}^{-1} \cdot \text{T}^{-1}$.

- b) In MHz/T: The gyromagnetic ratio of the expected magnetic moment of Dirac electron is $2.8007 \times 10^7 \text{MHz/T}$.

The gyromagnetic ration of the measured magnetic moment of Dirac electron is $2.803 \times 10^7 \text{MHz/T}$.

The gyromagnetic ratio of the proton(exptected) is $1.5277 \times 10^7 \text{MHz/T}$.

The gyromagnetic ratio of the proton(measured) is $4.26 \times 10^7 \text{MHz/T}$.

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The simplest reason is that the proton is not a point-like Dirac particle but a composite object made of quarks bound together by gluons. Its internal quark-gluon dynamics gives it a magnetic moment that deviates significantly from the value predicted by the naive Dirac equation.

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- a) in Figure 1

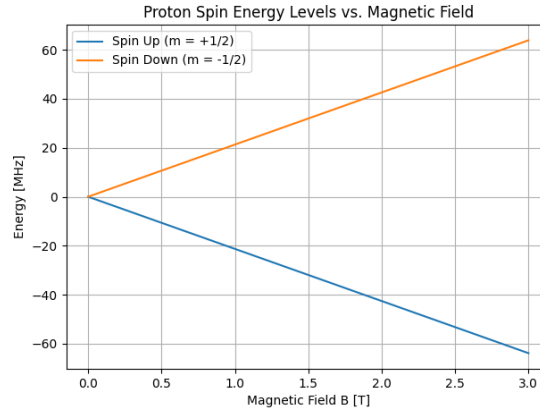


Figure 1: The energy level diagram for the spin-up and spin-down states of the proton as a function of the magnetic field.

- b)

$$\Delta E_Z = E_{\frac{1}{2}} - E_{-\frac{1}{2}} = \gamma \hbar B$$

- c) The dependency of ΔE_Z on magnetic field, in MHz/T: from

$$\Delta E_Z = \gamma \hbar B$$

we know, the dependency of ΔE_Z on magnetic field is $2.823 \times 10^{-26} \text{MHz/T}$

- d) let polarization be $P = \frac{N_{\uparrow} - N_{\downarrow}}{N_{\uparrow} + N_{\downarrow}}$.

Assume Boltzmann distribution, $P_i = \frac{1}{Z} \exp(-\frac{E_i}{k_B T})$, $P = \tanh \frac{\gamma \hbar B}{2 k_B T}$.

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Volume of the tube:

$$v_{tube} = \pi \times \left(\frac{d}{2}\right)^2 \times l = 0.393\text{mL}$$

volume of H_2O :

$$v_{H_2O} = 0.05 \times v_{tube} = 0.01965\text{mL}$$

a)

$$m_{H_2O} = 0.01965\text{g}$$

$$n_{H_2O} = \frac{0.01965\text{g}}{18\text{g/mol}} = 1.09 \times 10^{-3}\text{mol}$$

$$N_{H_2O} = n_{H_2O} \times N_A = 6.56 \times 10^{20}\text{molecules}$$

which means 1.312×10^{21} Hydrogen nuclei, that is 3.3×10^{21} Hydrogen nuclei per mL.

b) $N_H = 1.312 \times 10^{21}$

c) $2.18 \times 10^{-3}\text{mol}$

Larmor frequency, the rotating frame, and offset frequencies

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a) the number of protons in spin up/down states can be calculated as such:

$$\Delta N = PN_H = 1.367 \times 10^{16}$$

$$N_{\uparrow} = \frac{1}{2}(1 + P)N_H = 6.560 \times 10^{20}$$

$$N_{\downarrow} = \frac{1}{2}(1 - P)N_H = 6.559 \times 10^{20}$$

b) the net magnetic moment can be calculated as such:

$$\mu_{net} = \Delta N \mu_{proton} = 7.716 \times 10^{-11}$$

c) the order of magnitude of the magnetic field strength at 2cm is 10^{-12}T

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the shifts are calculated as such: $\delta = 10^6 \times \frac{\nu - \nu_{TMS}}{\nu_{TMS}}$, so in order to do get the physical value of the shifts, we do the revers of this calculation.

$$\Delta\delta = 3.80\text{ppm}$$

$$\Delta\nu = \Delta\delta \times 400.13 = 1520.494\text{Hz} \approx 9550\text{rads/s}$$

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The Larmor frequency is calculated as such:

$$\nu = -\frac{1}{2\pi}\gamma(1+\delta)B$$

using the conditions given in exercise, $\nu = -1.0072 \times 10^8 \text{Hz}$ or $-6.3249 \times 10^8 \text{rads/s}$

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What happened here is the RF pulses rotates the net magnetization away from z-axis. The angle of rotation can be calculated as such:

$$\theta = \omega_1 t = 2\pi\nu_1 t$$

From the data given, ν_1 can be calculated as

$$\nu_1 = \frac{\theta}{2\pi t} \approx 24.4 \text{kHz}$$

The 90 degree pulse length is $10.25\mu\text{s}$

As for the second null, it is clear that the magnetization has returned to +z.

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It does not. Here's why: let's first get the ω_1

$$\omega_1 = \frac{\frac{\pi}{2}}{\tau_{pulse}} \approx 25 \text{kHz}$$

The maximum offset here is:

$$\Delta = 45 \text{MHz} \approx 2.83 \times 10^8 \text{rads/s}$$

Notice $\Delta \gg \omega_1$, meaning the effective magnetic field is along the z-axis. The current magnetization can only tilt the spin by $\frac{\omega_1}{\Delta} \approx 1.4 \times 10^{-4}$ rads. Which is essentially negligible. Therefore it does not have a significant effect on C-13 nuclei

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Before the added pulses, the vector is along +y direction.

$\phi = +x$, the vector will be pointing at -z direction

$\phi = -x$, the vector will be pointing at +z direction

for both $\phi = -y$ and $\phi = +y$, the vector will be pointing at +y direction.